RISHANT PRAKASH

Ph.D. in Planetary and Space Sciences | Department of Physics

Doctoral student | University of Central Florida

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Research interests: Cosmochemistry; Solar System origins and volatile depletion; high- and low-temperature geochemistry; isotope measurements and mass spectrometry.

EDUCATION

University of Central Florida Aug 2026-Present
Ph.D. in Planetary and Space Sciences | Department of Physics

Arizona State University | School of Earth and Space Exploration Aug 2023-Jul 2026
M.S. in Geological Sciences; GPA: 3.9/4.0

Indian Institute of Science, Bengaluru 2021-2023
M.Tech. in Earth Sciences; GPA: 7.9/10

Hansraj College, University of Delhi 2019-2021
M.Sc. in Geology; GPA: 8.32/10; First Division with Distinction

Hansraj College, University of Delhi 2016-2019
B.Sc. in Geology; GPA: 7.46/10

RESEARCH EXPERIENCE

Titanium isotope anomalies in the solar nebula Started Aug 2023
Previous research, School of Earth and Space Exploration, Arizona State University

Advisor: Prof. Meenakshi Wadhwa

- Developed and refined in-situ LA-MC-ICP-MS protocols for high-precision Ti isotope measurements of diverse calcium-aluminum-rich inclusions (CAIs), including FUN CAIs.
- Synthesized Ti-doped CMAS glasses with varied V and Cr ratios as internal standards, using a gas-mixing furnace at the University of Arizona.
- Performed Mg purification in achondrites, including column chemistry, acid titration, and hazardous-waste management in Class 1000 cleanrooms.

Presolar grains and core-collapse supernovae Started Aug 2024
Previous research, School of Earth and Space Exploration, Arizona State University

Advisors: Prof. Patrick Young and Prof. Larry Nittler

- Used three-dimensional core-collapse supernova models to investigate Ti and Cr isotopic signatures measured in presolar oxide grains.
- Examined explosive nucleosynthesis, mixing, and fallback; compared simulated signatures with laboratory NanoSIMS datasets to investigate astrophysical source reservoirs.

Acid leaching, silicate weathering, and meteorite geochemistry 2021-2023
M.Tech. research, Centre for Earth Sciences, Indian Institute of Science, Bengaluru

Advisor: Prof. Ramananda Chakrabarti

- Conducted acid-leaching experiments on ultramafic to felsic silicate rocks, using non-HF acids and high-pressure asher digestion to investigate cation mobility.
- Examined silica content, residue weight, mineral stability, and major- and rare-earth-element behavior in leachates and residues.
- Characterized the Sadiya meteorite using bulk Ni, Co, and Fe data to investigate its H/L-chondrite classification.

Thesis

RESEARCH EXPERIENCE - CONTINUED

Amoeboid olivine aggregates in CM and CV chondrites May-Jul 2022

Summer research internship, Planetary Science Division, Physical Research Laboratory, Ahmedabad

Advisor: Dr. Dipak Panda

- Analyzed amoeboid olivine aggregates in Mukundpura and Allende chondrites using EDS and WDS on a JEOL 8530F field-emission electron-probe microanalyzer.
- Compared mineral chemistry and alteration histories, documenting phyllosilicates, carbonates, and poorly characterized phases in Mukundpura.
- Examined low-Ca pyroxene at aggregate margins to investigate nebular condensation and links with chondrule formation; presented the work at MetMESS 2022.

Project report

Land-use and land-cover change in Dehradun 2020-2021

M.Sc. dissertation, Department of Geology, University of Delhi

Advisor: Prof. Vimal Singh

- Used Landsat 7 ETM, Landsat 4-5 TM, and Landsat 8 OLI/TIRS imagery to classify land use and cover in 2001, 2011, and 2021.
- Applied maximum-likelihood supervised classification in ERDAS IMAGINE; assessed accuracy using error matrices and Kappa statistics and interpreted spatial and temporal changes.

Dissertation

Depositional environments in the Krishna-Godavari Basin 2019

Industry internship, Oil and Natural Gas Corporation, Dehradun

Advisor: GM Rajesh Sharma

- Integrated gamma-ray and resistivity logs, core observations, and biostratigraphy from four wells to interpret fluvial to marginal-marine synrift sequences.
- Analyzed sandstone maturity and diagenesis to assess reservoir quality and fluid mobility; used biostratigraphic and palynological evidence to establish Late Jurassic to Early Cretaceous ages.

Internship report

RESEARCH CONTRIBUTIONS

Prakash, R., Young, P. A., and Desch, S. J. *Three-dimensional core-collapse supernova clumps as sources of ^{54}Cr-rich presolar oxide nanograins.* 88th Annual Meeting of the Meteoritical Society, Frankfurt am Main, LPI Contribution 3162, abstract 5283. **Conference abstract; oral session.** Abstract 2026

Prakash, R., Putak Juricek, M., Spring, I., Rai, V., and Wadhwa, M. *Development of V- and Cr-doped Ti-CMAS glasses as standards for in situ titanium isotope analysis of calcium-aluminum-rich inclusions.* 88th Annual Meeting of the Meteoritical Society, Frankfurt am Main, LPI Contribution 3162, abstract 5278. **Conference abstract; poster session.** Abstract 2026

Prakash, R. and Young, P. *Presolar oxide grains as probes of symmetric and asymmetric core-collapse supernova explosions.* 247th American Astronomical Society Meeting, January 4-8, 2026, abstract 277.01. **Conference abstract.** Abstract 2026

Prakash, R. *Thermodynamic origins of ^{54}Cr-rich presolar oxide nanograins from three-dimensional core-collapse supernova models, with synthesis and characterization of Ti-CMAS glasses for in situ titanium isotope analysis.* Arizona State University. **Thesis.** Record 2026

Campbell, A., **Prakash, R.**, Chakraborty, A., Vierling, S., Garvie, L., and Wittmann, A. *Santa María del Oro (IIIAB, iron), Durango, Mexico.* Meteoritical Bulletin, No. 114; approved July 2025. **Classification contribution.** Database entry 2025

Prakash, R., Rai, V. K., Aigner, M., Hines, R., and Wadhwa, M. *Origin and distribution of Ti nucleosynthetic anomalies in the solar nebula.* SESE Symposium, February 22, 2025. **Poster.** 2025

Prakash, R. and Panda, D. *Geochemical analysis of amoeboid olivine aggregates in Mukundpura and Allende chondrites.* MetMESS, November 24-25, 2022, abstract S1-09. **Conference abstract and presentation.** Abstract 2022

AWARDS AND SCHOLASTIC ACHIEVEMENTS

- NASA Travel Award, 88th Meteoritical Society Annual Meeting, Frankfurt, Germany 2026
- SESE Exploration Graduate Fellowship, Arizona State University 2026
- Graduate College Travel Award, 247th American Astronomical Society Meeting 2026
- University Graduate Fellowship, College of Liberal Arts and Sciences, ASU 2025
- GATE Scholarship for M.Tech. study, Indian Institute of Science 2021-2023
- Summer Research Fellowship, Physical Research Laboratory, Ahmedabad 2022
- M.Sc. First Division with Distinction, Hansraj College, University of Delhi 2021
- Selected student volunteer, International Geological Congress 2020
- Second place, National Geological Quiz, AAPG Student Chapter, University of Delhi 2016-2017

GEOLOGICAL FIELD TRAINING

Meteor Crater and Sunset Crater, Arizona, USA Sep 2023
Studied overturned strata and impact-crater formation at Meteor Crater, aa lava flows at Sunset Crater, and distinctions between impact and fluid-interaction crater processes.

Ajeetgarh, Rajasthan, India Dec 2019
Mapped Precambrian geology and structural deformation; examined foliation, lineation, mineralization, and cross-bedding in the basal Alwar Group.

Dehradun-Mussoorie, Uttarakhand, India Sep 2018
Studied structural and sedimentological features, the Himalayan Frontal and Main Boundary thrusts, and the Ganga-Yamuna drainage divide; practiced geological mapping and field instruments.

Varanasi and the Vindhyan region, Uttar Pradesh, India Mar 2018
Collected Semri and Rewa formation samples, examined Proterozoic sedimentary structures, and measured stromatolites in the Sulkan Valley.

Nainital, Uttarakhand, India Dec 2016
Examined the geological setting of Nainital Lake and the structural and sedimentological characteristics of the Krol and Tal formations.

OUTREACH, LEADERSHIP, AND SERVICE

SESE Outreach, Buseck Center for Meteorite Studies, ASU From 2023
Shared extraterrestrial and terrestrial rocks with visitors from varied backgrounds as part of previous ASU outreach.

IISc Open Day, Centre for Earth Sciences 2022-2023
Introduced visitors to meteorites and science experiments.

Coordinator, AAPG Student Chapter, University of Delhi 2018-2019
Organized events and quiz competitions and managed chapter social media.

Technical Head, PIXELS Photography Society, Hansraj College 2017-2019
Led 40 undergraduates; organized photo walks, taught photojournalism and editing, and documented campus events.

Volunteer Teacher, NAV UMEED, North Delhi 2017-2019
Supported high-school students from marginalized backgrounds with mathematics and science in evening classes.

Technical Head, VISION Newspaper Society, Hansraj College 2017-2018
Led editorial design and publication of a biweekly student newspaper.

Volunteer, National Service Scheme Rural Development Wing 2016-2018
Facilitated workshops on personal hygiene, education, and poverty alleviation for marginalized youth in North Delhi.

TECHNICAL SKILLS

Analytical instruments: LA-MC-ICP-MS (Neptune; Photon Machine 193 nm UV laser); ICP-MS (Thermo X-Series II); EPMA (JEOL); SEM (Phenom XL).

Laboratory and experimental: Class 1000 and 10000 cleanrooms; column chemistry; acid digestion; gas-mixing furnace; high-pressure asher.

Programming: Python, C++, LaTeX.

Software: ArcGIS, ERDAS IMAGINE, OriginPro, Google Earth, Adobe Suite, Microsoft Office, Google Suite.

Languages: English and Hindi (fluent).